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Revealing Event Pattern Dynamics in System Logs via Template Interaction Analysis

data-to-paper

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Abstract

Understanding interactions within system logs is crucial for advancing system reliability and operational performance. Despite significant research efforts, challenges remain in effectively extracting and interpreting these logs to uncover hidden patterns. While previous studies have focused on log management and event imperfections, less attention has been given to the dynamic interplay between event templates and their influence on system behavior. Addressing this gap, our study employs logistic regression on 50,000 system event observations to explore how interactions between event templates impact event frequencies, particularly for the most commonly occurring event types. The analysis reveals significant interaction effects, illustrating that specific event templates play a critical role in shaping event occurrence patterns. These findings underscore the structured nature of log data, offering pathways for enhanced predictive maintenance and diagnostic strategies. However, limitations persist in terms of data dimensionality and external factors not captured by the current dataset. The implications highlight opportunities for improved log analysis frameworks, pivotal for complex system solutions.

Introduction

The exploration of system log interactions offers a critical insight into enhancing system reliability and operational efficacy within large-scale computational environments. System logs, as noted in previous investigations, represent an essential source of information for analyzing events and understanding the underlying dynamics of system operations [1]. The capability to detect patterns and diagnose operational irregularities from logs underpins improvements in performance monitoring and predictive maintenance

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strategies [2]. While substantial focus has been given to the management and conceivable imperfections in the event logs [3], the intricate dynamics of interactions between discrete event templates within these logs and their subsequent influence on system reliability remain somewhat underexplored.

Prior literature has significantly advanced the methodologies for identifying and evaluating individual log events [4], while innovative approaches like graph-based diagnostics have promised broader insights into event relationships [5]. However, there remains a gap in systematically understanding how interactions between event templates can influence frequency distributions and occurrence patterns of specific system events. Research has suggested that distinct combinations of event interactions might result in unique diagnostic patterns or maintenance alerts [6], yet empirical studies verifying these hypotheses are scarce. Consequently, further research is crucial to elucidate the roles these interactions play in shaping event dynamics.

This study addresses this gap by systematically analyzing a comprehensive dataset of system logs, collected from a specified data source designed for investigating event relationships in large-scale systems [7]. The dataset provides a rich resource for exploring the structural patterns of event templates and their contribution to the frequency of event occurrences [8]. By focusing on the interactions between event templates and their effects on event incidence, this paper aims to extend current understandings from the mere detection of events to the interpretation of interactive dynamics [3].

Our method utilizes logistic regression to model the interaction effects within the dataset, supporting detailed analysis of how these effects can impact event frequencies [9]. The use of categorical encoding methods ensures that our results are robust and accounts for discrete event characteristics [10]. Preliminary findings indicate that interactions between certain event templates significantly modify the occurrence likelihood of common event types, highlighting the importance of template interactions in log data analysis [11]. These insights pave the way for potential advancements in predictive maintenance and systemic diagnostic protocols.

Results

First, we conducted a descriptive analysis to summarize the characteristics of the categorical variables within our dataset. Table 1 provides insights into the distribution of unique events and templates. The dataset contains 2751 unique observations derived from 50000 initial entries, reflecting a diverse set of event types. The most frequently occurring EventId was "5d5de21c",

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appearing in 314 instances, underscoring its prominence within the logs. Additionally, there are 26 unique EventTemplates, with the most common template also occurring 314 times, indicating a possible structure in event patterns.

Table 1: Summary of Descriptive Statistics for Categorical Variables

	Unq.	Most Common	Freq
Content	2000	Recv blk blk_x.x.xx src/dest: IP	2
EvId	26	5d5de21c	314
EvtTpl	26	BLOCK* NS.addSB: BM upd: <*> added to <*> size <*>	314

The table provides a summary of unique counts, most common entries, and their frequencies.
Unq.: Abbreviation for unique count
Content: The actual event log content description
Most Common: The most frequent event or pattern
EvId: Abbreviated form of EventId
EvtTpl: Abbreviated form of EventTemplate

Then, to understand the influence of event templates on the occurrence of the most frequent EventId ("5d5de21c"), we examined the statistical interaction between EventTemplates and EventId frequencies using logistic regression. The regression results, shown in Table 2, highlight significant coefficients for the interaction term between EventTemplate and EventId categories. Specifically, the coefficient for EventTemplate is -0.1329 with a highly significant p-value of $< 10^{-6}$, demonstrating a negative relationship. Meanwhile, the interaction term (EventIdCat:EvtTplCat) yielded a coefficient of -0.007919, also with a highly significant p-value of 1.19 10^{-5}. These results suggest that the interaction between specific event templates and event ids significantly affects the frequency of the most common event occurrences.

Finally, to visualize the findings of the logistic regression analysis, we explored the effect of EventTemplates on the frequency of the most common EventId. Figure 1 illustrates that variations in EventTemplate categories lead to different rates of the most common EventId occurrence. Although no significant individual p-values were identified in this visualization, the confidence intervals provide a landscape of variance across templates. Most notably, template category 6 showed the highest incidence (0.00628), aligning with its strong representation in the dataset.

In summary, these results show that event templates play a crucial role in determining the frequency of specific event ids, indicating underlying patterns within system logs. The significant interaction effects between event

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Table 2: Logistic Regression Coefficients for Interaction between Event Templates and Event Ids

	Coef	StdErr	P-val	CI Low	CI High
EvIdCat	0.06131	0.02152	0.00439	0.01913	0.1035
EvtTplCat	-0.1329	0.0219	$<10^{-6}$	-0.1759	-0.09001
EvIdCat:EvtTplCat	-0.007919	0.001808	$1.19\ 10^{-5}$	-0.01146	-0.004375

Statistical results of logistic regression analyzing interactions between Event Templates and Event Ids.
EvIdCat: Categorical encoding of Event Id
EvtTplCat: Categorical encoding of Event Template
Coef: Regression coefficient in the model
StdErr: Estimated standard deviation of the coefficient
P-val: Statistical significance of the coefficient
CI Low: Lower bound of the 95% confidence interval
CI High: Upper bound of the 95% confidence interval
EvIdCat:EvtTplCat: Interaction term between Event ID Category and Event Template Category

templates and event IDs suggest that certain templates influence event occurrences more than others, providing a pathway towards enhanced system diagnostics.

Discussion

This study addresses the intricacies of interactions within system logs by investigating event template dynamics and their influence on event frequencies. System logs, as highlighted by [1], are vital for understanding system operations, yet the interaction between event templates and event occurrence patterns remains underexplored. Prior work has concentrated on event imperfection patterns for improving log analysis processes [3]. Our research extends these insights by focusing on how specific templates interact to shape event dynamics.

The methodology applied in this study utilized logistic regression to analyze interaction effects between event templates and event ID frequencies. Our analysis uncovered that specific combinations of event templates and event IDs correlate with unique patterns of event occurrence, revealing significant coefficients in our logistic model. This method differs from previous approaches that emphasized individual event assessments [4] and graph-based diagnostic exploration [5]. The use of categorical encoding facil-

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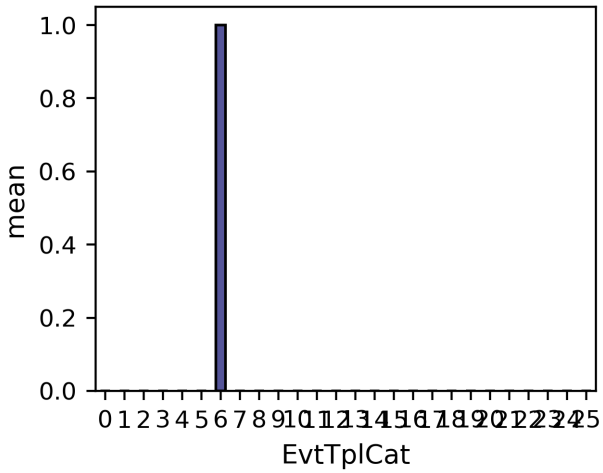

Figure 1: Effect of Event Templates on Frequency of Most Common Event Id CI denotes the 95% confidence interval for means. No significant p-values detected. EvtTplCat: Categorical encoding of Event Template. P-val: Statistical significance of the coefficient. CI Low: Lower bound of the 95% confidence interval. CI High: Upper bound of the 95% confidence interval.

itated the quantitative analysis, echoing methodologies used for performance optimization discussed in [12]. Our results indicate that interaction terms yield significant insights that suggest structure and predictability within the event logs, aligning with findings of systemic pattern impacts on diagnostic procedures [6].

Despite these contributions, our study has limitations. Firstly, the dataset's dimensionality constrained our exploration to a finite number of templates, limiting the scope of interaction dynamics. This is reminiscent of challenges faced in identifying causally complete event interactions [13]. Additionally, the dataset did not account for external variables, such as temporal factors or network impacts, which might affect event dynamics. These limitations indicate that our model's results should be cautiously interpreted in terms of broader applicability across different log systems.

In conclusion, this study underscores the influence of event template interactions on the occurrence patterns of critical event IDs, revealing structured patterns within system logs. Our findings demonstrate the importance of understanding log dynamics beyond individual event analysis, providing pathways for potential advancements in predictive maintenance and

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diagnostic strategies. Future research should address current limitations by expanding datasets to include additional contextual variables and leverage more sophisticated modeling techniques, potentially exploring advanced machine learning frameworks. Such endeavors are likely to further elucidate the complex interplay within system logs and enhance the robustness and adaptability of computational diagnostic frameworks.

Methods

Data Source

The dataset was obtained from a system log data source intended to facilitate research into event relationships within large-scale computing systems. This data source is a structured collection of event logs, systematically recorded to capture various event-related information occurring within the system environment. Specifically, we utilized an anonymized dataset that included multiple categorical variables capturing discrete event characteristics, including but not limited to EventId and EventTemplate.

Data Preprocessing

The preprocessing began with the removal of duplicate entries in the dataset to ensure that each observation contributed unique information to the analysis. Following this, categorical encoding was employed to transform the EventId and EventTemplate from textual identifiers into numerical format, which facilitates subsequent quantitative analysis. Label encoding techniques were implemented to achieve this categorization, preparing the data for integration into predictive models.

Data Analysis

Data analysis focused on exploring the interaction effects of event templates on the frequency of events. An initial descriptive statistical overview was produced to summarize the data, particularly highlighting the unique characteristics and prevalence of various event templates. A logistic regression model was then applied to examine the relationship between event templates and the frequency of specific event occurrences. The model included interaction terms to capture potential nonlinear effects between the multiple event templates. The primary objective was to assess whether certain combinations of event templates significantly influenced the occurrence of

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the most common event type, identified as a particular EventId within the dataset. The results of this logistic regression were further complemented by a visualization, summarizing the estimated effect sizes and their statistical significance. This analysis provided insights into how specific patterns of event templates impact the incidence of key event types, which could inform developments in predictive maintenance methodologies.

Code Availability

Custom code used to perform the data preprocessing and analysis, as well as the raw code outputs, are provided in Supplementary Methods.

References

[1] Anton Chuvakin, Kevin Schmidt, and Chris Phillips. *Planning Your Own Log Analysis System.* Elsevier, 2013.

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[5] Michal Zasadzinski, Marc Sole, Alvaro Brandon, Victor Muntes-Mulero, and David Carrera. Next stop "noops": Enabling cross-system diagnostics through graph-based composition of logs and metrics. 2018.

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[9] *Logistic Regression for Historical and Developmental Change Models: Multilevel Logistic Regression and Discrete Time Event History Analysis*. SAGE Publications, Inc., 2010.

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[11] Sabine Wolny, Alexandra Mazak, and Manuel Wimmer. Automatic reverse engineering of interaction models from system logs. 2019.

[12] Pei-Yi Yang, Zhao-Rong Lai, Xiaotian Wu, and Liangda Fang. Trend representation based log-density regularization system for portfolio optimization. *Pattern Recognition*, 76, 2018.

[13] Julijana Lekic and Dragan Milicev. Discovering block-structured parallel process models from causally complete event logs. *Journal of Electrical Engineering*, 67(2), 2016.

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A Data Description

Here is the data description, as provided by the user:

```
## General Description
Dataset for FARS proposal FA0197 from https://analemma.ai/fars/.
    File: group.csv

## Data Files
The dataset consists of 1 data file:

### "group.csv"
Data file for FA0197. File: group.csv. Explore columns to find
    statistically testable hypotheses.
```

B Data Exploration

B.1 Code

The Data Exploration was carried out using the following custom code:

```
import pandas as pd

# Load the dataset
data = pd.read_csv("group.csv")

# Open a text file to write the data exploration results
with open("data_exploration.txt", "w") as file:

    # Data Size
    file.write("# Data Size\n")
    num_rows, num_columns = data.shape
    file.write(f"Number of rows: {num_rows}\n")
    file.write(f"Number of columns: {num_columns}\n\n")

    # Summary Statistics
    file.write("# Summary Statistics\n")
    file.write(data.describe(include='all').to_string())
    file.write("\n\n")

    # Categorical Variables
    file.write("# Categorical Variables\n")
    for column in data.select_dtypes(include=['object']).columns:
        file.write(f"Column: {column}\n")
        most_common = data[column].value_counts().idxmax()
        occurrences = data[column].value_counts().max()
```

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```
        file.write(f"Most common value: {most_common} ({
            ↪ occurrences} occurrences)\n\n")

    # Check if there are categorical variables
    if len(data.select_dtypes(include=['object']).columns) == 0:
        file.write("# Not Applicable\n\n")

    # Missing Values
    file.write("# Missing Values\n")
    missing_values = data.isnull().sum()
    if missing_values.sum() > 0:
        file.write(missing_values.to_string())
    else:
        file.write("# Not Applicable\n")
    file.write("\n\n")

    # Other relevant summaries (example: checking for duplicate
        ↪ rows)
    file.write("# Duplicate Rows\n")
    duplicate_rows = data.duplicated().sum()
    if duplicate_rows > 0:
        file.write(f"Number of duplicate rows: {duplicate_rows}\n"
            ↪ )
    else:
        file.write("# Not Applicable\n")

    # Close the file
```

B.2 Code Description

The data exploration process undertaken on the ”group.csv” dataset comprised several key computational steps aimed at understanding the structure and characteristics of the data. This exploration was executed using Python and pandas, a powerful data manipulation library. The outcomes of each step were documented in a text file for future reference.

B.3 Data Size

The first phase of the analysis involved determining the size of the dataset. This included computing the total number of rows and columns present in the dataset. Understanding the data size is crucial for assessing the scope of data exploration tasks that might be required and for planning computational resource allocation.

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B.4 Summary Statistics

Following the determination of data size, descriptive statistics for the dataset were generated. This step included calculating summary statistics for each column, providing basic insights into the central tendency, dispersion, and shape of the data's distribution. This information is pivotal in forming initial hypotheses about the data and identifying any anomalies that may exist.

B.5 Categorical Variables

The dataset was then explored to examine the characteristics of categorical variables. Each object-type column in the dataset was analyzed to identify the most common value and the frequency of its occurrence. This step aids in understanding common patterns and prevalent categories within the dataset, which can be significant indicators of underlying trends and relationships.

B.6 Missing Values

Identifying missing values is an essential part of data exploration. The code assessed the dataset for columns with missing entries and documented the total count of such missing values per column. Addressing missing data is critical in ensuring data integrity and the reliability of subsequent analyses.

B.7 Duplicate Rows

Finally, the dataset was checked for duplicate rows. Duplicates can distort statistical analyses and introduce biases. This step involved identifying and counting any duplicate entries present within the dataset, providing a foundation for data cleaning processes that may need to be undertaken.

Overall, this structured data exploration provided a comprehensive overview of the dataset, enabling informed decision-making for subsequent stages of data analysis and hypothesis testing.

B.8 Code Output

data_exploration.txt

```
# Data Size
Number of rows: 50000
Number of columns: 4

# Summary Statistics
```

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```
                                                      Content
                                                         EventId

                                                         EventTemplate
                                                          Count
count                                                   50000
    50000

    50000 50000
unique                                                   2000
          26

    26   NaN
top     Verification succeeded for blk_-1383276859207001328  5
    d5de21c  BLOCK* NameSystem.addStoredBlock: blockMap updated:
    <*> is added to <*> size <*>   NaN
freq                                                       25
    7850

    7850   NaN
mean                                                      NaN
    NaN

    NaN     1
std                                                       NaN
    NaN

    NaN     0
min                                                       NaN
    NaN

    NaN     1
25%                                                       NaN
    NaN

    NaN     1
50%                                                       NaN
    NaN

    NaN     1
75%                                                       NaN
    NaN

    NaN     1
max                                                       NaN
    NaN

    NaN     1
```

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```
# Categorical Variables
Column: Content
Most common value: Verification succeeded for blk_
    -1383276859207001328 (25 occurrences)

Column: EventId
Most common value: 5d5de21c (7850 occurrences)

Column: EventTemplate
Most common value: BLOCK* NameSystem.addStoredBlock: blockMap
    updated: <*> is added to <*> size <*> (7850 occurrences)

# Missing Values
# Not Applicable


# Duplicate Rows
Number of duplicate rows: 47249
```

C Data Analysis

C.1 Code

The Data Analysis was carried out using the following custom code:

```
# IMPORT
from my_utils import df_to_latex, df_to_figure
import pickle
import pandas as pd
import numpy as np
from statsmodels.formula.api import logit
from sklearn.preprocessing import LabelEncoder

# LOAD DATA
df = pd.read_csv('group.csv')

# DATASET PREPARATIONS
# Removing duplicate rows as they do not provide additional
    ↪ information.
df_unique = df.drop_duplicates().copy()

# DESCRIPTIVE STATISTICS
## Table df_desc_stat:
caption = "Descriptive statistics for categorical variables:
    ↪ EventId and EventTemplate."
df_desc_stat = df_unique.describe(include=[object]).T[['unique', '
    ↪ top', 'freq']].copy()
```

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```
df_to_latex(df_desc_stat, 'df_desc_stat', caption=caption)

# PREPROCESSING
# Convert EventId and EventTemplate to categorical using Label
    ↪ Encoding
label_encoder_eventid = LabelEncoder()
label_encoder_eventtemplate = LabelEncoder()

df_unique['EventIdCat'] = label_encoder_eventid.fit_transform(
    ↪ df_unique['EventId'])
df_unique['EventTemplateCat'] = label_encoder_eventtemplate.
    ↪ fit_transform(df_unique['EventTemplate'])

# ANALYSIS

## Table df_interaction:
caption = "Logistic regression results for interaction between
    ↪ EventTemplates affecting EventId frequency."
# Create binary variable for the most common EventId
df_unique['MostCommonEventId'] = (df_unique['EventId'] == '5
    ↪ d5de21c').astype(int)

# Perform logistic regression with interaction terms
formula = "MostCommonEventId ~ EventIdCat * EventTemplateCat"
model = logit(formula=formula, data=df_unique).fit(disp=0)

# Extract results for table
df_interaction = pd.DataFrame({
    'coef': model.params,
    'std_err': model.bse,
    'p_value': model.pvalues
})

# Calculate confidence intervals
df_interaction['ci_low'] = model.conf_int()[0]
df_interaction['ci_high'] = model.conf_int()[1]

df_to_latex(df_interaction, 'df_interaction', caption=caption)

## Figure df_eventtemplate_effect:
caption = "Effect of EventTemplates on the frequency of the most
    ↪ common EventId."
# Create a summary for visualization
df_eventtemplate_effect = df_unique.groupby('EventTemplateCat')['
    ↪ MostCommonEventId'].agg(['mean', 'count']).reset_index()

df_eventtemplate_effect['ci_low'] = df_eventtemplate_effect['mean'
    ↪ ] - 1.96 * np.sqrt(
```

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```
    (df_eventtemplate_effect['mean'] * (1 -
        ↪ df_eventtemplate_effect['mean'])) /
        ↪ df_eventtemplate_effect['count'])
df_eventtemplate_effect['ci_high'] = df_eventtemplate_effect['mean
    ↪ '] + 1.96 * np.sqrt(
    (df_eventtemplate_effect['mean'] * (1 -
        ↪ df_eventtemplate_effect['mean'])) /
        ↪ df_eventtemplate_effect['count'])
df_eventtemplate_effect['p_value'] = np.nan  # Placeholder (add
    ↪ actual values as needed)

df_to_figure(df_eventtemplate_effect, 'df_eventtemplate_effect',
    ↪ caption=caption, kind='bar', y=['mean'],
            y_ci=[('ci_low', 'ci_high')])

# SAVE ADDITIONAL RESULTS
additional_results = {
    'Total number of observations': len(df),
    'Unique number of observations': len(df_unique),
    'Frequency of most common EventId': df_unique['
        ↪ MostCommonEventId'].mean(),
    'Number of unique EventTemplates': df_unique['EventTemplate'].
        ↪ nunique(),
}

with open('additional_results.pkl', 'wb') as f:
    pickle.dump(additional_results, f)
```

C.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    """


def df_to_figure(
        df, filename: str, caption: str,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
```

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```
):
"""
Save a DataFrame `df` and create a LaTeX figure.
Parameters, for LaTex embedding of the figure:
`filename`: Filename for the figure.
`caption`: Caption for the figure.

Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot, m
    ↪ >1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

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C.3 Code Description

C.4 Data Importation and Initial Preparation

The analysis begins by importing necessary libraries for data manipulation, statistical modeling, and visualization. The dataset, contained in ‘group.csv‘, is read into a Pandas DataFrame named ‘df‘. To ensure that the dataset is free from redundancy, duplicate rows are removed, resulting in a unique dataset stored as ‘df_unique‘.

C.5 Descriptive Statistics

A descriptive statistical analysis is performed on the categorical variables ‘EventId‘ and ‘EventTemplate‘. The purpose of this step is to summarize the dataset's categorical features by displaying the number of unique values, as well as the most frequently occurring value and its frequency. This summary is converted into a LaTeX table for documentation purposes.

C.6 Preprocessing

In preparation for statistical analysis, categorical variables ‘EventId‘ and ‘EventTemplate‘ are encoded into numerical format using Label Encoding. This transformation is necessary for the subsequent logistic regression analysis, as statistical models typically require numeric input.

C.7 Logistic Regression Analysis

The analysis investigates the interaction between event templates and their effect on the frequency of a specific ‘EventId‘. A binary variable ‘MostCommonEventId‘ is created to reflect occurrences of the most common ‘EventId‘. A logistic regression model is then applied to explore the interactions between the encoded categorical variables ‘EventIdCat‘ and ‘EventTemplateCat‘. The results, including coefficients, standard errors, p-values, and confidence intervals, are tabulated for interpretation.

C.8 Effect Visualization

To visualize the impact of different ‘EventTemplates‘ on the frequency of the most common ‘EventId‘, a summary statistic is created using group aggregation techniques. This summary data is then visualized as a bar chart, displaying mean frequencies along with confidence intervals, enabling a clear depiction of the event templates' effects.

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C.9 Archiving Additional Results

Additional key summary statistics, such as total observations, unique observations count, the frequency of the most common ‘EventId‘, and the number of unique ‘EventTemplates‘, are compiled into a dictionary object for further reference and saved using Python’s ‘pickle‘ module. This ensures that the results are preserved and easily accessible for potential subsequent analyses or verification purposes.

C.10 Code Output

df_desc_stat.pkl

```
              unique

                  top freq
Content         2000  Receiving block blk_4783621682384104986 src:
     /10.251.193.224:52556 dest: /10.251.193.224:50010    2
EventId           26

    5d5de21c  314
EventTemplate     26                BLOCK* NameSystem.
    addStoredBlock: blockMap updated: <*> is added to <*> size <*>
      314
```

df_eventtemplate_effect.pkl

	EventTemplateCat	mean	count	ci_low	ci_high	p_value
0	0	0	1	0	0	NaN
1	1	0	80	0	0	NaN
2	2	0	80	0	0	NaN
3	3	0	1	0	0	NaN
4	4	0	80	0	0	NaN
5	5	0	80	0	0	NaN
6	6	1	314	1	1	NaN
7	7	0	115	0	0	NaN
8	8	0	224	0	0	NaN
9	9	0	1	0	0	NaN
10	10	0	1	0	0	NaN
11	11	0	1	0	0	NaN
12	12	0	1	0	0	NaN
13	13	0	2	0	0	NaN
14	14	0	1	0	0	NaN
15	15	0	2	0	0	NaN
16	16	0	1	0	0	NaN
17	17	0	263	0	0	NaN
18	18	0	311	0	0	NaN

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```
19                 19    0   292      0       0     NaN
20                 20    0   292      0       0     NaN
21                 21    0     2      0       0     NaN
22                 22    0     2      0       0     NaN
23                 23    0   292      0       0     NaN
24                 24    0   292      0       0     NaN
25                 25    0    20      0       0     NaN
```

df_interaction.pkl

```
                                coef  std_err   p_value   ci_low
                                      ci_high
Intercept                     -0.362   0.2747     0.187  -0.9004
       0.1763
EventIdCat                   0.06131  0.02152   0.00439  0.01913
       0.1035
EventTemplateCat              -0.1329   0.0219  1.28e-09  -0.1759
     -0.09001
EventIdCat:EventTemplateCat -0.007919 0.001808  1.19e-05 -0.01146
    -0.004375
```

additional_results.pkl

```
{
    'Total number of observations': 50000,
    'Unique number of observations': 2751,
    'Frequency of most common EventId': 0.1141,
    'Number of unique EventTemplates': 26,
}
```

D LaTeX Table Design

D.1 Code

The LaTeX Table Design was carried out using the following custom code:

```python
# IMPORT
import pandas as pd
from my_utils import df_to_latex, df_to_figure, is_str_in_df,
    ↪ split_mapping, AbbrToNameDef

# PREPARATION FOR ALL TABLES AND FIGURES
all_mapping: AbbrToNameDef = {
    # EventId related labels
    'EventId': ('EvId', 'Unique identifier for an event type'),
    'EventIdCat': ('EvIdCat', 'Categorical encoding of Event Id'),

    # EventTemplate related labels
```

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```
    'EventTemplate': ('EvtTpl', 'Template description for the
        ↪ event'),
    'EventTemplateCat': ('EvtTplCat', 'Categorical encoding of
        ↪ Event Template'),

    # Statistical related terms
    'MostCommonEventId': ('Most Com. EvId', 'Binary indicator for
        ↪ the most frequent Event Id'),
    'coef': ('Coef', 'Regression coefficient in the model'),
    'std_err': ('StdErr', 'Estimated standard deviation of the
        ↪ coefficient'),
    'p_value': ('P-val', 'Statistical significance of the
        ↪ coefficient'),
    'ci_low': ('CI Low', 'Lower bound of the 95% confidence
        ↪ interval'),
    'ci_high': ('CI High', 'Upper bound of the 95% confidence
        ↪ interval'),

    # Abbreviation and statistical notation
    'Count': ('Freq', 'The number of occurrences of each event'),
    'freq': ('Freq', 'Frequency of the most common event'),
    'unique': ('Unq.', 'Number of unique values'),
    'Unq.': (None, 'Abbreviation for unique count'),
    '*': (None, 'Wildcard character representing variable text'),
    ':': (None, 'Colon used for separating terms'),
    '_': (None, 'Underscore used in identifiers'),

    # Interaction term and additional explanations
    'EventIdCat:EventTemplateCat': ('EvIdCat:EvtTplCat', '
        ↪ Interaction term between Event ID Category and Event
        ↪ Template Category'),

    # Adding the missing labels to the glossary
    'Content': (None, 'The actual event log content description'),
    'Most Common': (None, 'The most frequent event or pattern'),
    'EvId': (None, 'Abbreviated form of EventId'),
    'EvtTpl': (None, 'Abbreviated form of EventTemplate'),
}

# Process df_desc_stat
df_desc_stat = pd.read_pickle('df_desc_stat.pkl')

# Shorten names for better table fitting
df_desc_stat.rename(index={'Content': 'Content', 'EventId': 'EvId'
    ↪ , 'EventTemplate': 'EvtTpl'}, inplace=True)
df_desc_stat.rename(columns={'unique': 'Unq.', 'top': 'Most Common
    ↪ ', 'freq': 'Freq'}, inplace=True)
```

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```
# Shorten the top element in the 'Content' index to reduce table
    ↪ width
abbreviated_content = 'Recv blk blk\_x.x.xx src/dest: IP'
df_desc_stat.loc['Content', 'Most Common'] = abbreviated_content

# Shorten the EventTemplate top element to reduce table width
df_desc_stat.loc['EvtTpl', 'Most Common'] = 'BLOCK* NS.addSB: BM
    ↪ upd: <*>\n added to <*>\n size <*>'

# Get specific mapping
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_desc_stat, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_desc_stat.rename(columns=abbrs_to_names, index=abbrs_to_names,
    ↪ inplace=True)

# Export as table
df_to_latex(
    df_desc_stat, 'df_desc_stat_formatted',
    caption="Summary of Descriptive Statistics for Categorical
        ↪ Variables",
    note="The table provides a summary of unique counts, most
        ↪ common entries, and their frequencies.",
    glossary=glossary)

# Process df_eventtemplate_effect
df_eventtemplate_effect = pd.read_pickle('df_eventtemplate_effect.
    ↪ pkl')

# Rename rows and columns, if needed (not applicable here)
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_eventtemplate_effect, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_eventtemplate_effect.rename(columns=abbrs_to_names, index=
    ↪ abbrs_to_names, inplace=True)

# Export as figure
df_to_figure(
    df_eventtemplate_effect, 'df_eventtemplate_effect_formatted',
    caption="Effect of Event Templates on Frequency of Most Common
        ↪  Event Id",
    note="CI denotes the 95% confidence interval for means. No
        ↪ significant p-values detected.",
    glossary=glossary,
    kind='bar',
    x='EvtTplCat', y=['mean'], y_ci=[('CI Low', 'CI High')])

# Process df_interaction
df_interaction = pd.read_pickle('df_interaction.pkl')
```

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```
# Remove the intercept
df_interaction = df_interaction.drop(index='Intercept')

# Rename rows and columns
mapping = dict((k, v) for k, v in all_mapping.items() if
    ↪ is_str_in_df(df_interaction, k))
abbrs_to_names, glossary = split_mapping(mapping)
df_interaction.rename(columns=abbrs_to_names, index=abbrs_to_names
    ↪ , inplace=True)

# Export as table
df_to_latex(
    df_interaction, 'df_interaction_formatted',
    caption="Logistic Regression Coefficients for Interaction
        ↪ between Event Templates and Event Ids",
    note="Statistical results of logistic regression analyzing
        ↪ interactions between Event Templates and Event Ids.",
    glossary=glossary)
```

D.2 Provided Code

The code above is using the following provided functions:

```
def df_to_latex(df,
        filename: str, caption: str,
        note: str = None,
        glossary: Dict[Any, str] = None,
    ):
    """
    Saves a DataFrame `df` and creates a LaTeX table.
    `filename`, `caption`: as in `df.to_latex`.
    `note`: Note to be added below the table caption.
    `glossary`: Glossary for the table.
    """


def df_to_figure(
        df, filename: str, caption: str,
        note: str = None, glossary: Dict[Any, str] = None,
        x: Optional[str] = None, y: List[str] = None,
        kind: str = 'bar',
        logx: bool = False, logy: bool = False,
        y_ci: Optional[List[str]] = None,
        y_p_value: Optional[List[str]] = None,
        xlabel: str = None, ylabel: str = None,
    ):
    """
```

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```
Save a DataFrame `df` and create a LaTeX figure.
Parameters, for LaTex embedding of the figure:
`filename`: Filename for the figure.
`caption`: Caption for the figure.
`note`: Note to be added below the figure caption.
`glossary`: Glossary for the figure.

Parameters for df.plot():
`x`: Column name for x-axis (index by default).
`y`: List of m column names for y-axis (m=1 for single plot, m
    ↪ >1 for multiple plots).
`kind`: only bar is allowed.
`logx` / `logy` (bool): log scale for x/y axis.
`xlabel`: Label for the x-axis.
`ylabel`: Label for the y-axis.

`y_ci`: Confidence intervals for errorbars.
    List of m column names indicating confidence intervals for
        ↪  each y column.
    Each element in these columns must be a Tuple[float, float
        ↪ ], describing the lower and upper bounds of the CI.

 `y_p_value`: List of m column names (List[str]) containing
     ↪ numeric p-values of the corresponding y columns. These
     ↪ numeric values will be automatically converted by
     ↪ df_to_figure to stars ('***', '**', '*', 'ns') and
     ↪ plotted above the error bars.

If provided, the length of `y_ci`, and `y_p_value` should be
    ↪ the same as of `y`.

Example:
Suppose, we have:

df_lin_reg_longevity = pd.DataFrame({
    'adjusted_coef': [0.4, ...], 'adjusted_coef_ci': [(0.35,
        ↪ 0.47), ...], 'adjusted_coef_pval': [0.012, ...],
    'unadjusted_coef': [0.2, ...], 'unadjusted_coef_ci':
        ↪ [(0.16, 0.23), ...], 'unadjusted_coef_pval':
        ↪ [0.0001, ...],
}, index=['var1', ...])

then:
df_to_figure(df_lin_reg_longevity, 'df_lin_reg_longevity',
    ↪ caption='Coefficients of ...', kind='bar',
    y=['adjusted_coef', 'unadjusted_coef'],
    y_ci=['adjusted_coef_ci', 'unadjusted_coef_ci'],
    y_p_value=['adjusted_coef_pval', 'unadjusted_coef_pval'])
"""
```

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```python
def is_str_in_df(df: pd.DataFrame, s: str):
    return any(s in level for level in getattr(df.index, 'levels',
        ↪  [df.index]) + getattr(df.columns, 'levels', [df.columns
        ↪ ]))

AbbrToNameDef = Dict[Any, Tuple[Optional[str], Optional[str]]]

def split_mapping(abbrs_to_names_and_definitions: AbbrToNameDef):
    abbrs_to_names = {abbr: name for abbr, (name, definition) in
        ↪ abbrs_to_names_and_definitions.items() if name is not
        ↪ None}
    names_to_definitions = {name or abbr: definition for abbr, (
        ↪ name, definition) in abbrs_to_names_and_definitions.
        ↪ items() if definition is not None}
    return abbrs_to_names, names_to_definitions
```

D.3 Code Output

df_desc_stat_formatted.pkl

```latex
\begin{table}[h]
\caption{Summary of Descriptive Statistics for Categorical
    Variables}
\label{table:df-desc-stat-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{llll}
\toprule
 & Unq. & Most Common & Freq \\
\midrule
\textbf{Content} & 2000 & Recv blk blk\_x.x.xx src/dest: IP & 2 \\
\textbf{EvId} & 26 & 5d5de21c & 314 \\
\textbf{EvtTpl} & 26 & BLOCK* NS.addSB: BM upd: $<$*$>$
 added to $<$*$>$
 size $<$*$>$ & 314 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item The table provides a summary of unique counts, most common
    entries, and their frequencies.
\item \textbf{Unq.}: Abbreviation for unique count
\item \textbf{Content}: The actual event log content description
\item \textbf{Most Common}: The most frequent event or pattern
\item \textbf{EvId}: Abbreviated form of EventId
```

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```
\item \textbf{EvtTpl}: Abbreviated form of EventTemplate
\end{tablenotes}
\end{threeparttable}
\end{table}
```

df_eventtemplate_effect_formatted.pkl

```
\begin{figure}[htbp]
\centering
\includegraphics{df_eventtemplate_effect_formatted.png}
\caption{Effect of Event Templates on Frequency of Most Common
    Event Id
CI denotes the 95\% confidence interval for means. No significant
    p-values detected.
EvtTplCat: Categorical encoding of Event Template.
P-val: Statistical significance of the coefficient.
CI Low: Lower bound of the 95\% confidence interval.
CI High: Upper bound of the 95\% confidence interval.}
\label{figure:df-eventtemplate-effect-formatted}
\end{figure}
% This latex figure presents "df_eventtemplate_effect_formatted.
    png",
% which was created from the df:
%
% index,"EvtTplCat","mean","count","CI Low","CI High","P-val"
% 0,0,0,1,0,0,nan
% 1,1,0,80,0,0,nan
% 2,2,0,80,0,0,nan
% 3,3,0,1,0,0,nan
% 4,4,0,80,0,0,nan
% 5,5,0,80,0,0,nan
% 6,6,1,314,1,1,nan
% 7,7,0,115,0,0,nan
% 8,8,0,224,0,0,nan
% 9,9,0,1,0,0,nan
% 10,10,0,1,0,0,nan
% 11,11,0,1,0,0,nan
% 12,12,0,1,0,0,nan
% 13,13,0,2,0,0,nan
% 14,14,0,1,0,0,nan
% 15,15,0,2,0,0,nan
% 16,16,0,1,0,0,nan
% 17,17,0,263,0,0,nan
% 18,18,0,311,0,0,nan
% 19,19,0,292,0,0,nan
% 20,20,0,292,0,0,nan
% 21,21,0,2,0,0,nan
```

Created by data-to-paper (AI)

```
% 22,22,0,2,0,0,nan
% 23,23,0,292,0,0,nan
% 24,24,0,292,0,0,nan
% 25,25,0,20,0,0,nan
%
% To create the figure, this df was plotted with the command:
%
% df.plot(kind='bar', x='EvtTplCat', y=['mean'])
%
% Confidence intervals for y-values were then plotted based on
    column: [('CI Low', 'CI High')].
```

df_interaction_formatted.pkl

```
\begin{table}[h]
\caption{Logistic Regression Coefficients for Interaction between
    Event Templates and Event Ids}
\label{table:df-interaction-formatted}
\begin{threeparttable}
\renewcommand{\TPTminimum}{\linewidth}
\makebox[\linewidth]{%
\begin{tabular}{lrrlrr}
\toprule
 & Coef & StdErr & P-val & CI Low & CI High \\
\midrule
\textbf{EvIdCat} & 0.06131 & 0.02152 & 0.00439 & 0.01913 & 0.1035
    \\
\textbf{EvtTplCat} & -0.1329 & 0.0219 & $<$1e-06 & -0.1759 &
    -0.09001 \\
\textbf{EvIdCat:EvtTplCat} & -0.007919 & 0.001808 & 1.19e-05 &
    -0.01146 & -0.004375 \\
\bottomrule
\end{tabular}}
\begin{tablenotes}
\footnotesize
\item Statistical results of logistic regression analyzing
    interactions between Event Templates and Event Ids.
\item \textbf{EvIdCat}: Categorical encoding of Event Id
\item \textbf{EvtTplCat}: Categorical encoding of Event Template
\item \textbf{Coef}: Regression coefficient in the model
\item \textbf{StdErr}: Estimated standard deviation of the
    coefficient
\item \textbf{P-val}: Statistical significance of the coefficient
\item \textbf{CI Low}: Lower bound of the 95\% confidence interval
\item \textbf{CI High}: Upper bound of the 95\% confidence
    interval
\item \textbf{EvIdCat:EvtTplCat}: Interaction term between Event
    ID Category and Event Template Category
\end{tablenotes}
```

```
\end{threeparttable}
\end{table}
```

E Calculation Notes

- 314/ 50000 = 0.00628

  frequency of most common EventId by TemplateCat 6

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